In [1]:
```
import torch
image=torch.randn(1,3,224,224)
print(image.shape)
print(image.dtype)
print(image.device)
```

```
torch.Size([1, 3, 224, 224])
torch.float32
cpu
```

In [3]:
```
import torch
import torch.nn as nn
conv=nn.Conv2d(
    in_channels=3,
    out_channels=16,
    kernel_size=3,
    stride=1,
    padding=1
)
```

In [5]:
```
import torch
import torch.nn as nn
relu=nn.ReLU()
x=torch.tensor([-2.0,-1.0,0.0,1.0,2.0])
print(relu(x))
```

```
tensor([0., 0., 0., 1., 2.])
```

In [6]:
```
import torch.nn.functional as F
print(F.relu(x))
```

```
tensor([0., 0., 0., 1., 2.])
```

In [7]:
```
import torch
import torch.nn as nn
pool=nn.MaxPool2d(2)
x=torch.randn(1,3,32,32)
output=pool(x)
print(output.shape)
```

```
torch.Size([1, 3, 16, 16])
```

In [12]:
```
import torch
import torch.nn as nn
class CNN(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1=nn.Conv2d(3,32,3,padding=1)
        self.conv2=nn.Conv2d(32,64,3,padding=1)
        self.pool=nn.MaxPool2d(2)
        self.fc1=nn.Linear(64*8*8,128)
        self.fc2=nn.Linear(128,10)
        self.relu=nn.ReLU()
    def forward(self,x):
        x=self.pool(self.relu(self.conv1(x)))
        x=self.pool(self.relu(self.conv2(x)))
```

```
        x=x.view(x.size(0),-1
                )
        x=self.relu(self.fc1(x))
        x=self.fc2(x)
        return x
model=CNN()
print(model)
dummy=torch.randn(4,3,32,32)
out=model(dummy)
print("Output shape:",out.shape)
```

```
CNN(
  (conv1): Conv2d(3, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (conv2): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
  (pool): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  (fc1): Linear(in_features=4096, out_features=128, bias=True)
  (fc2): Linear(in_features=128, out_features=10, bias=True)
  (relu): ReLU()
)
Output shape: torch.Size([4, 10])
```

In [13]:

```
total = sum(p.numel() for p in model.parameters() if p.requires_grad)
print(f"Total trainable parameters: {total:,}")
for name, param in model.named_parameters():
    print(f"{name:20s}  {str(list(param.shape)):25s}  {param.numel():>8,}")
```

```
Total trainable parameters: 545,098
conv1.weight          [32, 3, 3, 3]                   864
conv1.bias            [32]                             32
conv2.weight          [64, 32, 3, 3]               18,432
conv2.bias            [64]                             64
fc1.weight            [128, 4096]                 524,288
fc1.bias              [128]                           128
fc2.weight            [10, 128]                     1,280
fc2.bias              [10]                             10
```

In [14]:

```
import torchvision
import torchvision.transforms as transforms
from torch.utils.data import DataLoader
transform = transforms.Compose([
    transforms.ToTensor()
])
trainset = torchvision.datasets.CIFAR10(
    root='./data',
    train=True,
    download=True,
    transform=transform
)
testset = torchvision.datasets.CIFAR10(
    root='./data',
    train=False,
    download=True,
    transform=transform
)
trainloader = DataLoader(trainset, batch_size=64, shuffle=True,  num_workers=2)
testloader  = DataLoader(testset,  batch_size=64, shuffle=False, num_workers=2)
```

```
print(f"Training samples : {len(trainset):,}")
print(f"Test samples     : {len(testset):,}")
print(f"Classes          : {trainset.classes}")
```

```
Downloading https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz to ./data/cifar-
10-python.tar.gz
100%|████████████████████████████████████████
█| 170M/170M [00:16<00:00, 10.3MB/s]
Extracting ./data/cifar-10-python.tar.gz to ./data
Files already downloaded and verified
Training samples : 50,000
Test samples     : 10,000
Classes          : ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog', 'frog',
'horse', 'ship', 'truck']
```

In [19]:

```
!pip uninstall -y tensorflow numpy ml-dtypes keras
!pip install numpy==1.26.4
!pip install tensorflow==2.17.0
```

```
Found existing installation: tensorflow 2.21.0
Uninstalling tensorflow-2.21.0:
  Successfully uninstalled tensorflow-2.21.0
Found existing installation: numpy 2.2.6
Uninstalling numpy-2.2.6:
  Successfully uninstalled numpy-2.2.6
Found existing installation: ml_dtypes 0.5.4
Uninstalling ml_dtypes-0.5.4:
  Successfully uninstalled ml_dtypes-0.5.4
Found existing installation: keras 3.12.2
Uninstalling keras-3.12.2:
  Successfully uninstalled keras-3.12.2
WARNING: Running pip as the 'root' user can result in broken permissions and conflic
ting behaviour with the system package manager, possibly rendering your system unusa
ble.It is recommended to use a virtual environment instead: https://pip.pypa.io/warn
ings/venv. Use the --root-user-action option if you know what you are doing and want
to suppress this warning.
Looking in indexes: https://pypi.org/simple, https://pypi.ngc.nvidia.com
Collecting numpy==1.26.4
  Downloading numpy-1.26.4-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.wh
l.metadata (61 kB)
Downloading numpy-1.26.4-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.whl
(18.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 18.2/18.2 MB 189.2 MB/s eta 0:00:00
Installing collected packages: numpy
Successfully installed numpy-1.26.4
WARNING: Running pip as the 'root' user can result in broken permissions and conflic
ting behaviour with the system package manager, possibly rendering your system unusa
ble.It is recommended to use a virtual environment instead: https://pip.pypa.io/warn
ings/venv. Use the --root-user-action option if you know what you are doing and want
to suppress this warning.

[notice] A new release of pip is available: 24.2 -> 26.1.2
[notice] To update, run: python -m pip install --upgrade pip
Looking in indexes: https://pypi.org/simple, https://pypi.ngc.nvidia.com
Collecting tensorflow==2.17.0
  Downloading tensorflow-2.17.0-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_
64.whl.metadata (4.2 kB)
Requirement already satisfied: absl-py>=1.0.0 in /usr/local/lib/python3.10/dist-pack
ages (from tensorflow==2.17.0) (2.1.0)
Requirement already satisfied: astunparse>=1.6.0 in /usr/local/lib/python3.10/dist-p
ackages (from tensorflow==2.17.0) (1.6.3)
Requirement already satisfied: flatbuffers>=24.3.25 in /usr/local/lib/python3.10/dis
t-packages (from tensorflow==2.17.0) (25.12.19)
Requirement already satisfied: gast!=0.5.0,!=0.5.1,!=0.5.2,>=0.2.1 in /usr/local/li
b/python3.10/dist-packages (from tensorflow==2.17.0) (0.6.0)
Requirement already satisfied: google-pasta>=0.1.1 in /usr/local/lib/python3.10/dist
-packages (from tensorflow==2.17.0) (0.2.0)
Requirement already satisfied: h5py>=3.10.0 in /usr/local/lib/python3.10/dist-packag
es (from tensorflow==2.17.0) (3.14.0)
Requirement already satisfied: libclang>=13.0.0 in /usr/local/lib/python3.10/dist-pa
ckages (from tensorflow==2.17.0) (18.1.1)
Collecting ml-dtypes<0.5.0,>=0.3.1 (from tensorflow==2.17.0)
  Downloading ml_dtypes-0.4.1-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_6
4.whl.metadata (20 kB)
Requirement already satisfied: opt-einsum>=2.3.2 in /usr/local/lib/python3.10/dist-p
```

```
ackages/opt_einsum-3.4.0-py3.10.egg (from tensorflow==2.17.0) (3.4.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.10/dist-packages
(from tensorflow==2.17.0) (23.2)
Collecting protobuf!=4.21.0,!=4.21.1,!=4.21.2,!=4.21.3,!=4.21.4,!=4.21.5,<5.0.0dev,>
=3.20.3 (from tensorflow==2.17.0)
  Downloading protobuf-4.25.9-cp37-abi3-manylinux2014_x86_64.whl.metadata (541 byte
s)
Requirement already satisfied: requests<3,>=2.21.0 in /usr/local/lib/python3.10/dist
-packages (from tensorflow==2.17.0) (2.32.3)
Requirement already satisfied: setuptools in /usr/local/lib/python3.10/dist-packages
(from tensorflow==2.17.0) (70.3.0)
Requirement already satisfied: six>=1.12.0 in /usr/local/lib/python3.10/dist-package
s (from tensorflow==2.17.0) (1.16.0)
Requirement already satisfied: termcolor>=1.1.0 in /usr/local/lib/python3.10/dist-pa
ckages (from tensorflow==2.17.0) (3.3.0)
Requirement already satisfied: typing-extensions>=3.6.6 in /usr/local/lib/python3.1
0/dist-packages (from tensorflow==2.17.0) (4.12.2)
Requirement already satisfied: wrapt>=1.11.0 in /usr/local/lib/python3.10/dist-packa
ges (from tensorflow==2.17.0) (1.16.0)
Requirement already satisfied: grpcio<2.0,>=1.24.3 in /usr/local/lib/python3.10/dist
-packages (from tensorflow==2.17.0) (1.62.1)
Collecting tensorboard<2.18,>=2.17 (from tensorflow==2.17.0)
  Downloading tensorboard-2.17.1-py3-none-any.whl.metadata (1.6 kB)
Collecting keras>=3.2.0 (from tensorflow==2.17.0)
  Downloading keras-3.12.2-py3-none-any.whl.metadata (5.9 kB)
Collecting tensorflow-io-gcs-filesystem>=0.23.1 (from tensorflow==2.17.0)
  Downloading tensorflow_io_gcs_filesystem-0.37.1-cp310-cp310-manylinux_2_17_x86_64.
manylinux2014_x86_64.whl.metadata (14 kB)
Requirement already satisfied: numpy<2.0.0,>=1.23.5 in /usr/local/lib/python3.10/dis
t-packages (from tensorflow==2.17.0) (1.26.4)
Requirement already satisfied: wheel<1.0,>=0.23.0 in /usr/local/lib/python3.10/dist-
packages (from astunparse>=1.6.0->tensorflow==2.17.0) (0.44.0)
Requirement already satisfied: rich in /usr/local/lib/python3.10/dist-packages (from
keras>=3.2.0->tensorflow==2.17.0) (13.7.1)
Requirement already satisfied: namex in /usr/local/lib/python3.10/dist-packages (fro
m keras>=3.2.0->tensorflow==2.17.0) (0.1.0)
Requirement already satisfied: optree in /usr/local/lib/python3.10/dist-packages (fr
om keras>=3.2.0->tensorflow==2.17.0) (0.13.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.1
0/dist-packages (from requests<3,>=2.21.0->tensorflow==2.17.0) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.10/dist-packag
es (from requests<3,>=2.21.0->tensorflow==2.17.0) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.10/dist-
packages (from requests<3,>=2.21.0->tensorflow==2.17.0) (2.0.7)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.10/dist-
packages (from requests<3,>=2.21.0->tensorflow==2.17.0) (2024.8.30)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.10/dist-pac
kages (from tensorboard<2.18,>=2.17->tensorflow==2.17.0) (3.7)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/l
ib/python3.10/dist-packages (from tensorboard<2.18,>=2.17->tensorflow==2.17.0) (0.7.
2)
Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.10/dist-pac
kages (from tensorboard<2.18,>=2.17->tensorflow==2.17.0) (3.0.4)
Requirement already satisfied: MarkupSafe>=2.1.1 in /usr/local/lib/python3.10/dist-p
ackages (from werkzeug>=1.0.1->tensorboard<2.18,>=2.17->tensorflow==2.17.0) (2.1.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.10/di
```

```
st-packages (from rich->keras>=3.2.0->tensorflow==2.17.0) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.10/
dist-packages (from rich->keras>=3.2.0->tensorflow==2.17.0) (2.18.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.10/dist-packages
(from markdown-it-py>=2.2.0->rich->keras>=3.2.0->tensorflow==2.17.0) (0.1.2)
Downloading tensorflow-2.17.0-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_6
4.whl (601.3 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 601.3/601.3 MB 19.5 MB/s eta 0:00:0000:0
100:01
Downloading keras-3.12.2-py3-none-any.whl (1.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.5/1.5 MB 138.7 MB/s eta 0:00:00
Downloading ml_dtypes-0.4.1-cp310-cp310-manylinux_2_17_x86_64.manylinux2014_x86_64.w
hl (2.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 2.2/2.2 MB 163.8 MB/s eta 0:00:00
Downloading protobuf-4.25.9-cp37-abi3-manylinux2014_x86_64.whl (295 kB)
Downloading tensorboard-2.17.1-py3-none-any.whl (5.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.5/5.5 MB 143.1 MB/s eta 0:00:00
Downloading tensorflow_io_gcs_filesystem-0.37.1-cp310-cp310-manylinux_2_17_x86_64.ma
nylinux2014_x86_64.whl (5.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 5.1/5.1 MB 164.4 MB/s eta 0:00:00
Installing collected packages: tensorflow-io-gcs-filesystem, protobuf, ml-dtypes, te
nsorboard, keras, tensorflow
  Attempting uninstall: protobuf
    Found existing installation: protobuf 7.35.0
    Uninstalling protobuf-7.35.0:
      Successfully uninstalled protobuf-7.35.0
  Attempting uninstall: tensorboard
    Found existing installation: tensorboard 2.16.2
    Uninstalling tensorboard-2.16.2:
      Successfully uninstalled tensorboard-2.16.2
Successfully installed keras-3.12.2 ml-dtypes-0.4.1 protobuf-4.25.9 tensorboard-2.1
7.1 tensorflow-2.17.0 tensorflow-io-gcs-filesystem-0.37.1
WARNING: Running pip as the 'root' user can result in broken permissions and conflic
ting behaviour with the system package manager, possibly rendering your system unusa
ble.It is recommended to use a virtual environment instead: https://pip.pypa.io/warn
ings/venv. Use the --root-user-action option if you know what you are doing and want
to suppress this warning.

[notice] A new release of pip is available: 24.2 -> 26.1.2
[notice] To update, run: python -m pip install --upgrade pip
```

In [1]:

```python
import tensorflow as tf
print("TensorFlow Version:", tf.__version__)
```

```
2026-06-04 05:59:25.201405: I tensorflow/core/util/port.cc:153] oneDNN custom operat
ions are on. You may see slightly different numerical results due to floating-point
round-off errors from different computation orders. To turn them off, set the enviro
nment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-06-04 05:59:25.211175: E external/local_xla/xla/stream_executor/cuda/cuda_fft.c
c:485] Unable to register cuFFT factory: Attempting to register factory for plugin c
uFFT when one has already been registered
2026-06-04 05:59:25.223517: E external/local_xla/xla/stream_executor/cuda/cuda_dnn.c
c:8454] Unable to register cuDNN factory: Attempting to register factory for plugin
cuDNN when one has already been registered
2026-06-04 05:59:25.227031: E external/local_xla/xla/stream_executor/cuda/cuda_blas.
cc:1452] Unable to register cuBLAS factory: Attempting to register factory for plugi
n cuBLAS when one has already been registered
2026-06-04 05:59:25.236022: I tensorflow/core/platform/cpu_feature_guard.cc:210] Thi
s TensorFlow binary is optimized to use available CPU instructions in performance-cr
itical operations.
To enable the following instructions: AVX2 AVX512F AVX512_VNNI AVX512_BF16 AVX512_FP
16 AVX_VNNI AMX_TILE AMX_INT8 AMX_BF16 FMA, in other operations, rebuild TensorFlow
with the appropriate compiler flags.
2026-06-04 05:59:26.208612: W tensorflow/compiler/tf2tensorrt/utils/py_utils.cc:38]
TF-TRT Warning: Could not find TensorRT
TensorFlow Version: 2.17.0
```

In [2]:

```
import os

os.environ['TF_CPP_MIN_LOG_LEVEL'] = '3'

import tensorflow as tf
from tensorflow.keras import layers, models

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

x_train = x_train.reshape(-1, 28, 28, 1).astype("float32") / 255.0
x_test = x_test.reshape(-1, 28, 28, 1).astype("float32") / 255.0

model = models.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Conv2D(16, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(10, activation='softmax')
])

model.summary()

conv_params = model.layers[0].count_params()

print("Trainable parameters in first Conv2D layer:", conv_params)
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 26, 26, 16)	160
max_pooling2d_1 (MaxPooling2D)	(None, 13, 13, 16)	0
flatten_1 (Flatten)	(None, 2704)	0
dense_1 (Dense)	(None, 10)	27,050

Total params: 27,210 (106.29 KB)

Trainable params: 27,210 (106.29 KB)

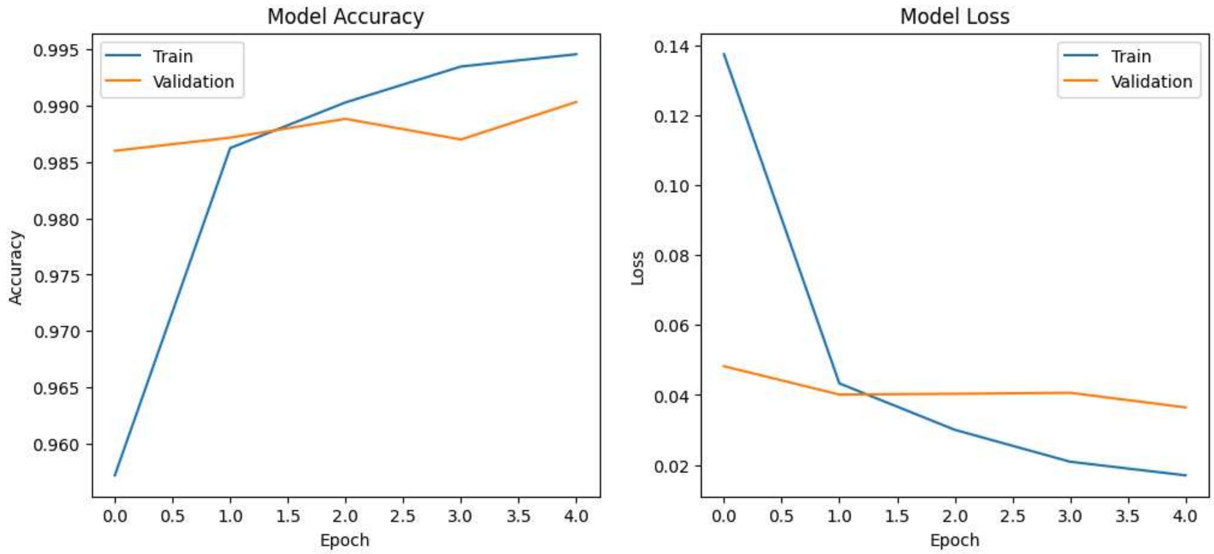
Non-trainable params: 0 (0.00 B)

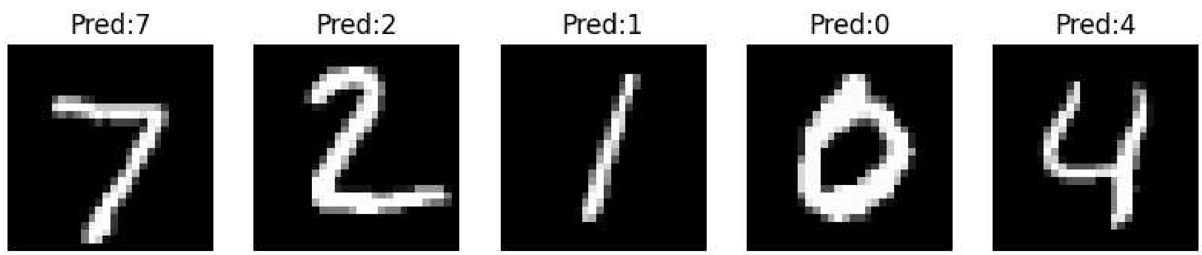
Trainable parameters in first Conv2D layer: 160

In [3]:
```
import os
os.environ['TF_CPP_MIN_LOG_LEVEL'] = '3'

import tensorflow as tf
import matplotlib.pyplot as plt
from tensorflow.keras import layers, models

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data()

x_train = x_train.reshape(-1, 28, 28, 1).astype("float32") / 255.0
x_test = x_test.reshape(-1, 28, 28, 1).astype("float32") / 255.0

model = models.Sequential([
    layers.Input(shape=(28, 28, 1)),
    layers.Conv2D(32, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Conv2D(64, (3, 3), activation='relu'),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    x_train,
    y_train,
    epochs=5,
    validation_split=0.1,
    verbose=1
)

test_loss, test_acc = model.evaluate(x_test, y_test)
```

```python
print("Test Accuracy:", test_acc)
print("Test Loss:", test_loss)

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend(['Train', 'Validation'])

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend(['Train', 'Validation'])

plt.show()

plt.figure(figsize=(10, 4))

for i in range(5):
    plt.subplot(1, 5, i + 1)
    plt.imshow(x_test[i].reshape(28, 28), cmap='gray')
    prediction = model.predict(x_test[i:i+1], verbose=0)
    plt.title(f"Pred:{prediction.argmax()}")
    plt.axis('off')

plt.show()
```

```
Epoch 1/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 53s 30ms/step - accuracy: 0.9572 - loss: 0.1375 - val
_accuracy: 0.9860 - val_loss: 0.0482
Epoch 2/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 51s 30ms/step - accuracy: 0.9862 - loss: 0.0433 - val
_accuracy: 0.9872 - val_loss: 0.0401
Epoch 3/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 51s 30ms/step - accuracy: 0.9903 - loss: 0.0301 - val
_accuracy: 0.9888 - val_loss: 0.0404
Epoch 4/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 51s 30ms/step - accuracy: 0.9935 - loss: 0.0209 - val
_accuracy: 0.9870 - val_loss: 0.0406
Epoch 5/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━ 51s 30ms/step - accuracy: 0.9946 - loss: 0.0170 - val
_accuracy: 0.9903 - val_loss: 0.0365
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 11ms/step - accuracy: 0.9894 - loss: 0.0332
Test Accuracy: 0.9894000291824341
Test Loss: 0.03324095904827118
```


Model Accuracy
Train
Validation
Accuracy
0.995
0.990
0.985
0.980
0.975
0.970
0.965
0.960
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Epoch
Model Loss
Train
Validation
Loss
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Epoch


Pred:7
Pred:2
Pred:1
Pred:0
Pred:4

In [ ]: